



ALLERGIC, INFLAMMATORY, AND IMMUNOLOGIC (RHEUMATIC) DISORDERS

MODULE LEARNING OBJECTIVES

Describe the nursing management of a client with an allergic disorder.

Use the nursing process as a framework for care of the client with an inflammatory rheumatic disorder.

ALLERGIC DISORDERS

TYPES OF HYPERSENSITIVITY REACTIONS

<i>Hypersensitivity Reaction</i>	<i>Description</i>
Type I <u>IgE-mediated; quick onset after exposure</u> Allergic	Bee stings Latex Certain medications (e.g. Penicillin)
Type II <u>Cytotoxic/antibody-mediated</u> Cytotoxic	Hemolytic reactions Goodpasture syndrome Hyperacute graft rejection
Type III <u>Immune complex/IgG/IgM mediated</u> Immune complex deposition	Hypersensitivity pneumonitis Systemic lupus erythematosus Polyarteritis nodosa Serum sickness
Type IV <u>Delayed or cell-mediated</u> Delayed	Chronic graft rejections PPD test Latex Nickel Poison ivy

ALLERGIC REACTIONS

- Allergy
- Allergen
- Atopy

BACKGROUND

- * **COEXISTENCE of ATOPIC DERMATITIS, ASTHMA, & ALLERGIES like ALLERGIC RHINITIS & CONJUNCTIVITIS**
- ~ INDIVIDUALS with ONE ATOPIC CONDITION at ↑↑ RISK of DEVELOPING ANOTHER ONE
- ~ CAN PERSIST for YEARS or DECADES



DIAGNOSIS

- * SKIN TESTING
- * ALLERGEN-SPECIFIC IgE TESTING
- * PRESENTATION & FAMILY HISTORY



TREATMENT

- * **ECZEMA**
 - ~ FREQUENT MOISTURIZING
 - ~ TOPICAL STEROIDS
 - ~ TOPICAL CALCINEURIN INHIBITORS
- * **ASTHMA**
 - ~ SHORT-ACTING BETA AGONIST
 - ~ INHALED & ORAL CORTICOSTEROIDS
 - ~ NEBULIZED BRONCHODILATORS
- * **ALLERGIC RHINITIS**
 - ~ INTRANASAL ANTIHISTAMINE SPRAYS
 - ~ CORTICOSTEROIDS
 - ~ NASAL IRRIGATION
 - ~ ANTIHISTAMINE EYE DROPS
 - ~ COOL COMPRESSES
 - ~ ARTIFICIAL TEARS



SIGNS & SYMPTOMS

- * **ECZEMA**
 - ~ RASHES or SCALY PATCHES
- * **ASTHMA**
 - ~ CHRONIC INFLAMMATION, NARROWING of AIRWAYS, & DIFFICULTY BREATHING
- * **ALLERGIC RHINITIS**
 - ~ SNEEZING, RHINORRHEA, & NASAL ITCHING

CAUSES

- * **GENETIC PREDISPOSITION for ENHANCED IMMUNE RESPONSE against SPECIFIC ALLERGENS**
- * **ENVIRONMENTAL TRIGGERS**
 - ~ ANIMAL DANDER, LATEX, POLLEN, INSECT STINGS
- * **VIRAL INFECTIONS**
- * **FOODS**
 - ~ NUTS, SHELLFISH, EGGS



ANSWER: ALLERGY, ALLERGEN, ATOPY

- **Allergy**

- An exaggerated, inappropriate immune response to a substance that is harmless to most people – a type I (IgE-mediated) hypersensitivity in most cases, ranging from a runny nose to anaphylaxis.

- **Allergen**

- The antigen that triggers the allergic response: foods (peanut, shellfish, egg, milk), drugs (penicillin, sulfa, NSAIDs, contrast dye), insect venom, latex, pollen, dust mites, animal dander, mold.

- **Atopy**

- The inherited tendency to make IgE against common environmental allergens – shows up as the “atopic triad”: allergic rhinitis, asthma and atopic dermatitis (eczema), often running in families.

EVALUATION OF ALLERGIC DISORDERS

Describe how the nurse evaluates a client's allergy status. What questions are appropriate to ask and why?

ANSWER: EVALUATING A CLIENT'S ALLERGY STATUS (1 OF 2)

- **Ask, and why**

- “What are you allergic to?” – drugs, foods, latex, environmental, insect stings, contrast dye; include over-the-counter products and herbals.
- “What exactly happens when you are exposed?” – separates a true allergy (hives, swelling, wheeze, anaphylaxis) from a side effect or intolerance (nausea from codeine, upset stomach from antibiotics); only true reactions go on the allergy band.
- “How soon after exposure, and how severe? Did you need epinephrine or an ER visit?” – guides risk and whether an EpiPen is needed.
- “When did it last happen, and what treated it?” – some drug allergies are outgrown; documents what worked.
- “Any family history of allergy, asthma or eczema?” – atopy runs in families. “Any latex reactions, or allergies to banana, kiwi, avocado, chestnut?” – latex-fruit cross-reactivity.
- “Do you have asthma or take beta blockers?” – both make anaphylaxis harder to treat.

ANSWER: EVALUATING A CLIENT'S ALLERGY STATUS (2 OF 2)

- **Then**

- Physical assessment (skin, eyes, nose, lungs), review of skin-prick or serum IgE (RAST) results, eosinophil count; apply the red allergy band, document in the chart and the pharmacy profile, and re-check before every new medication, contrast study or surgery.



**Potentially
life-
threatening
allergic
reaction**



**Rapid release
of IgE-
mediated
chemicals like
histamine**



**Mild,
moderate, or
severe
symptoms**



**Symptoms are
sudden in
onset and
may progress
over minutes
or hours**



**Penicillin is
most common
pharmacologi
c cause**

ANAPHYLAXIS

ANAPHYLAXIS CAUSES AND MANIFESTATIONS

- Food
- Medications
- Insect stings
- Latex



- Bronchoconstriction
- Angioedema
- Urticaria
- Hypotension





Assess client, including vital signs



Priority medication is IM epinephrine



Oxygen



Adjunct therapies

Antihistamines
Corticosteroids
Bronchodilators
IV fluids

ANAPHYLAXIS: TREATMENT

SELF- ADMINISTRATION OF EPINEPHRINE

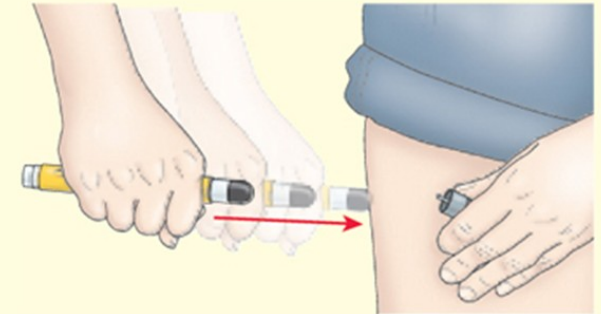
Self-Administration of Epinephrine

The nurse instructs the patient to:

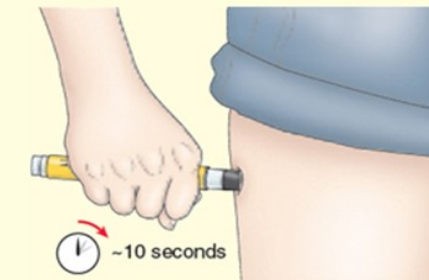
1. After removing the autoinjector from its carrying tube, grasp the unit with the orange tip (injecting end) pointing downward. Form a fist around the unit with the orange tip down; with your other hand, remove the blue safety release cap.



2. Hold the black tip near outer thigh. Swing and jab firmly into the outer thigh until a click is heard with the device perpendicular (90-degree angle) to the thigh. Do NOT inject into buttocks.



Hold firmly against the thigh for approximately 10 seconds. Remove the unit from the thigh, and gently massage the injection area for 10 seconds. Call 911 and seek immediate medical attention. Carefully place the used autoinjector unit, needle-end first, into the container without bending the needle. Screw on the storage tube completely, and take it with you to the hospital.



URTICARIA (HIVES)

- Pinkish, edematous elevations (wheals of variable size)
- Associated with itching
- Lesions come and go in minutes to hours
- Acute urticaria from an allergen is not an emergency like anaphylaxis
- Antihistamines are used for itching
- If it persists for longer than 6 weeks, it is called chronic urticaria, and often no cause is found



FOOD ALLERGIES



- 5% of adults have food allergies
- Genetic predisposition
- Can occur with any food
- Can result in simple urticaria or severe anaphylaxis
- If allergy results in anaphylaxis, will need an Epi-Pen at home/person
- May be hidden in unsuspected foods

LATEX ALLERGY

- 4 - 7% of general population
- At-risk populations
- Powder in latex gloves becomes carrier for latex proteins; Can become airborne and inhaled during removal
- Common cause of anaphylaxis is during the intraoperative period

How can healthcare workers decrease their exposure to latex?

ANSWER: REDUCING LATEX EXPOSURE FOR HEALTHCARE WORKERS

- Use non-latex gloves (nitrile, vinyl) for routine tasks and powder-free, low-protein latex gloves only when latex is truly needed – powder carries the protein into the air.
- Avoid oil-based hand creams under latex gloves (they break the glove down and increase protein release); wash hands with mild soap after removing gloves.
- Know the latex-containing supplies – catheters, tourniquets, blood-pressure cuffs, adhesive tape, syringe stoppers, some IV ports – and choose latex-free alternatives; keep latex-free carts.
- Recognize early signs in yourself: itchy, red, cracked hands, hives, runny nose, watery eyes, wheeze after wearing gloves – report to employee health and get tested.
- Anyone with a latex allergy: latex-free environment, allergy band and chart alert, tell every provider, carry epinephrine, and note the cross-reacting foods (banana, avocado, kiwi, chestnut).

RHEUMATIC DISORDERS

RHEUMATIC DISEASE

- Rheumatic diseases include more than 100 different disorders
- Three distinct characteristics
 - Inflammation (redness, swelling and tenderness)
 - Autoimmunity
 - Degeneration and destruction of joints
- Primarily affect joints, but also muscles, bone, ligament, tendons, cartilage
- Can affect a single joint or multiple joints

RHEUMATOID ARTHRITIS



1992



2002

Wiesenhutter

- Chronic systemic inflammatory disease (autoimmune)
- Destruction of connective tissue and synovial membrane within the joint
- Weakens the joint, leading to dislocation and permanent deformity
- Bilateral and symmetric
- Typically begins in small joints of hands and feet

RHEUMATOID ARTHRITIS: CLINICAL MANIFESTATIONS

- Inflammation, tenderness, and stiffness of the joints
- Moderate to severe pain with morning stiffness lasting longer than 30 minutes
- Joint deformities, muscle atrophy, and decreased range of motion in the affected joint
- Low grade fever, fatigue, and weakness
- Joints are spongy on palpation (filled with fluid)
- Client may have systemic manifestations

Rheumatoid Arthritis (Late stage)

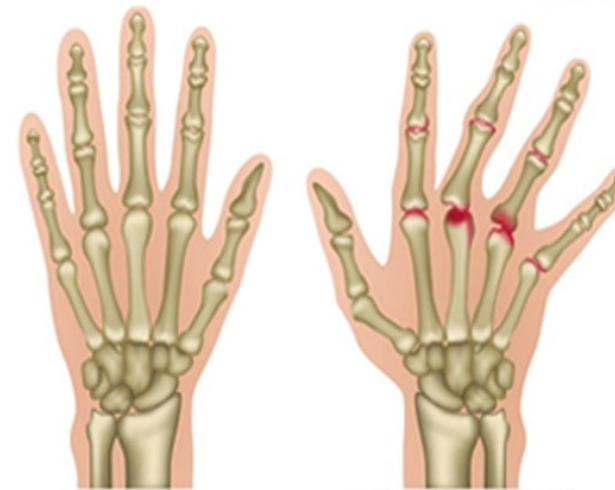
Boutonniere
deformity
of thumb

Ulnar deviation of
metacarpophalangeal
joints

Swan-neck deformity
of fingers



Rheumatoid nodules



RHEUMATOID NODULES

- Occur more often in clients with elevated serum rheumatoid factor
- Nontender and moveable lump of tissue, often over extensor surfaces like the elbow
- Often associated with rapidly progressive and destructive disease
- Can be injected with cortisone or surgically removed if very bothersome



RHEUMATOID ARTHRITIS: DIAGNOSTICS

- Serum rheumatoid factor (RF) antibody
- Erythrocyte sedimentation rate (ESR)
- C-reactive protein (CRP)
- Radiography



Figure 1

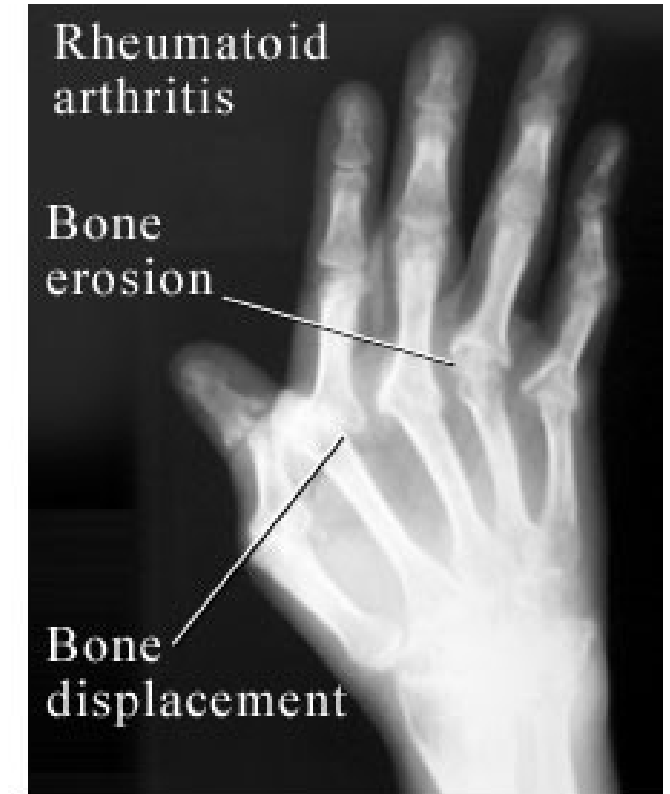
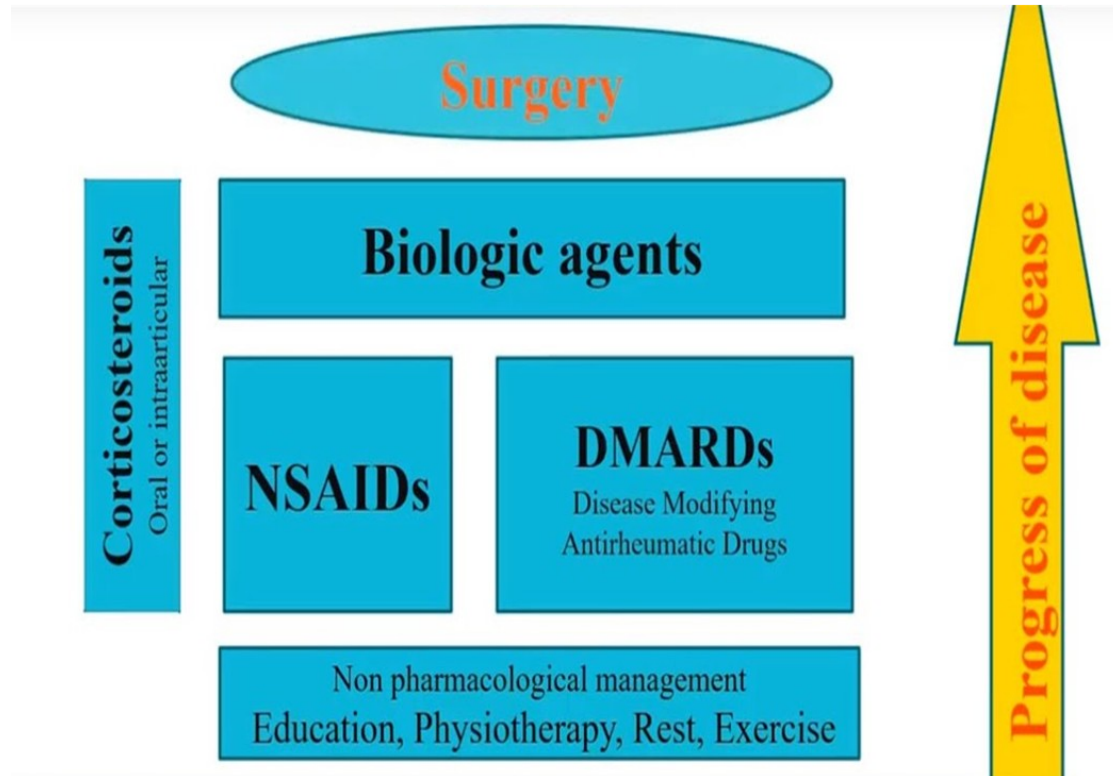


Figure 2

RHEUMATOID ARTHRITIS: MEDICAL MANAGEMENT



- NSAIDs
- DMARDs
- Corticosteroids
- Reconstructive surgery
- Joint arthroplasty

RHEUMATOID ARTHRITIS: PHARMACOTHERAPY

- Nonsteroidal anti-inflammatory drugs (NSAIDs)
- Non-biologic DMARD: Methotrexate
- Biologic DMARD: Infliximab

Describe pharmacotherapy education the nurse should provide to the client with rheumatoid arthritis.

ANSWER: RHEUMATOID ARTHRITIS PHARMACOTHERAPY EDUCATION (1 OF 2)

- **NSAIDs (ibuprofen, naproxen, celecoxib)**
 - Take with food; watch for GI bleeding (black stools, abdominal pain), kidney problems and raised blood pressure; do not combine NSAIDs; they relieve pain and inflammation but do not slow the disease.
- **Methotrexate (non-biologic DMARD)**
 - Once a WEEK, never daily – a daily dose is fatal; take folic acid as prescribed; avoid alcohol; report mouth sores, fever, cough or shortness of breath, bruising, jaundice; labs (CBC, liver, kidney) every 4–12 weeks; strict contraception for both partners – it causes birth defects; takes 6–8 weeks to work.

ANSWER: RHEUMATOID ARTHRITIS PHARMACOTHERAPY EDUCATION (2 OF 2)

- **Infliximab and other biologic DMARDs (etanercept, adalimumab)**

- Suppress the immune system: TB test and hepatitis screen before starting; report any fever, cough or infection at once and hold the dose when sick; no live vaccines; infusion reactions with infliximab – premedicate and monitor during and after; teach injection technique and refrigeration for self-injected biologics.

- **Corticosteroids (prednisone)**

- Lowest dose for the shortest time; take in the morning with food; never stop suddenly; watch glucose, blood pressure, weight, mood, bone density (calcium and vitamin D), and infection.

- **General**

- Keep taking DMARDs even when you feel well; keep all lab appointments; combine with rest, heat and cold, splints, and exercise.

RHEUMATOID ARTHRITIS: NURSING MANAGEMENT

Assist with and encourage physical activity

Balance activity and rest

Provide referrals for OT and PT to maintain maximal function and ADLs

Facilitate use of assistive devices

Monitor for medication effectiveness

Adequate nutrition

Nonpharmacologic pain relief

Follow routine health screenings

Report manifestations that can indicate RA exacerbation

Express feelings regarding body image

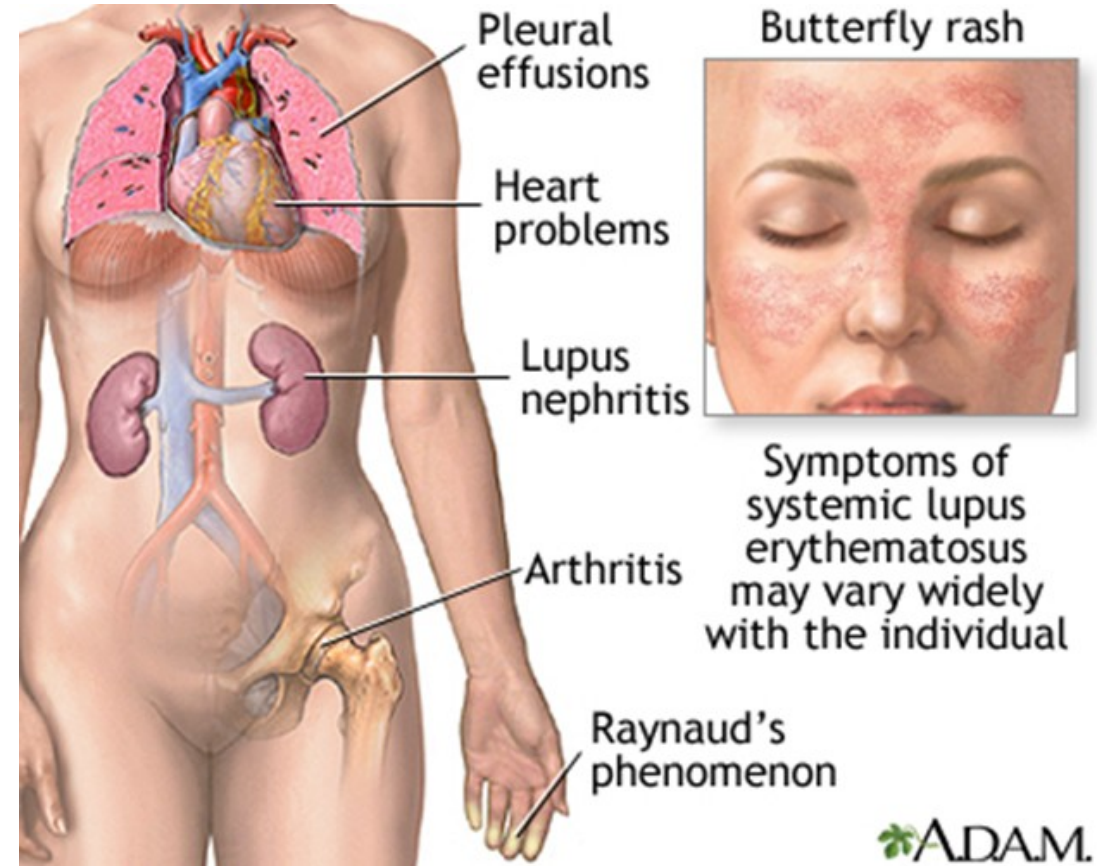
SYSTEMIC LUPUS ERYTHEMATOSUS

SYSTEMIC LUPUS ERYTHEMATOSU S (SLE)

- Chronic, progressive, systemic inflammatory disease that can cause failure of major organs and body systems
- Exaggerated production of autoantibodies that attack many sites in the body
- Immunologic, hormonal, and environmental factors
- > 80% of clients with lupus are female
- 3 times more common in African Americans than Caucasians
- Varies in severity and progression
- Periods of exacerbations and remissions
- No cure

SLE: CLINICAL MANIFESTATIONS

- Clinical manifestations caused by inflammation due to autoantibody attack
- Systemic
- Musculoskeletal
- Cutaneous
- Cardiovascular
- Renal
 - Nephritis is the most common cause of increased mortality in SLE
- Neurological



SLE: DIAGNOSTICS

H&P

Serum diagnostics

- Antinuclear antibodies
 - Present in autoimmune disorders and certain cancers
 - Positive in 95% of SLE cases
- ESR and CRP elevated
- CBC may show pancytopenia

SLE: MEDICAL MANAGEMENT

- Goals of medical management:
 - Symptom control
 - Prevention of further organ damage
- Pharmacotherapy
 - NSAIDs
 - Corticosteroids
 - Biologic DMARDs
 - Non-biologic DMARDs
 - Antimalarials

Describe pharmacotherapy education the nurse should provide to the client with SLE.

ANSWER: SLE PHARMACOTHERAPY EDUCATION (1 OF 2)

- **Hydroxychloroquine (antimalarial)**

- The backbone drug for almost every client – keeps flares away and protects organs; take with food; takes months to work; eye exam at baseline and yearly for retinal toxicity; report vision changes; generally safe in pregnancy.

- **Corticosteroids**

- For flares and organ involvement; morning dose with food; never stop abruptly; watch glucose, blood pressure, weight, mood, infection, bone loss (calcium, vitamin D) and avascular necrosis of the hip.

- **NSAIDs**

- For joint pain and fever; with food; avoid in lupus nephritis and kidney disease; watch for GI bleeding.

ANSWER: SLE PHARMACOTHERAPY

EDUCATION (2 OF 2)

- **Immunosuppressants (azathioprine, mycophenolate, cyclophosphamide, methotrexate)**
 - For nephritis and severe disease: regular CBC, liver and kidney labs; report infection, bleeding, bruising; strict contraception – most are teratogenic; cyclophosphamide needs hydration and bladder protection.
- **Biologics (belimumab, anifrolumab)**
 - Infusion or injection; screen for infection first; no live vaccines; report fever, depression or suicidal thoughts.
- **For everyone**
 - Sun protection every day (SPF 30+, hats, cover up) because UV triggers flares; keep vaccines current (inactivated); rest during flares; report fever, chest pain, edema, foamy urine or new rash promptly; plan pregnancy only in remission and with the rheumatologist.

SLE: NURSING MANAGEMENT

Focused
assessment

Monitor for
medication
effectiveness

Monitor for
adverse effects of
medication

Adequate nutrition

Pain relief:
pharmacologic &
nonpharmacologic

Follow routine
health screenings

Report
manifestations
that can indicate
SLE exacerbation
or complications

Teach to avoid
cold for
hands/toes

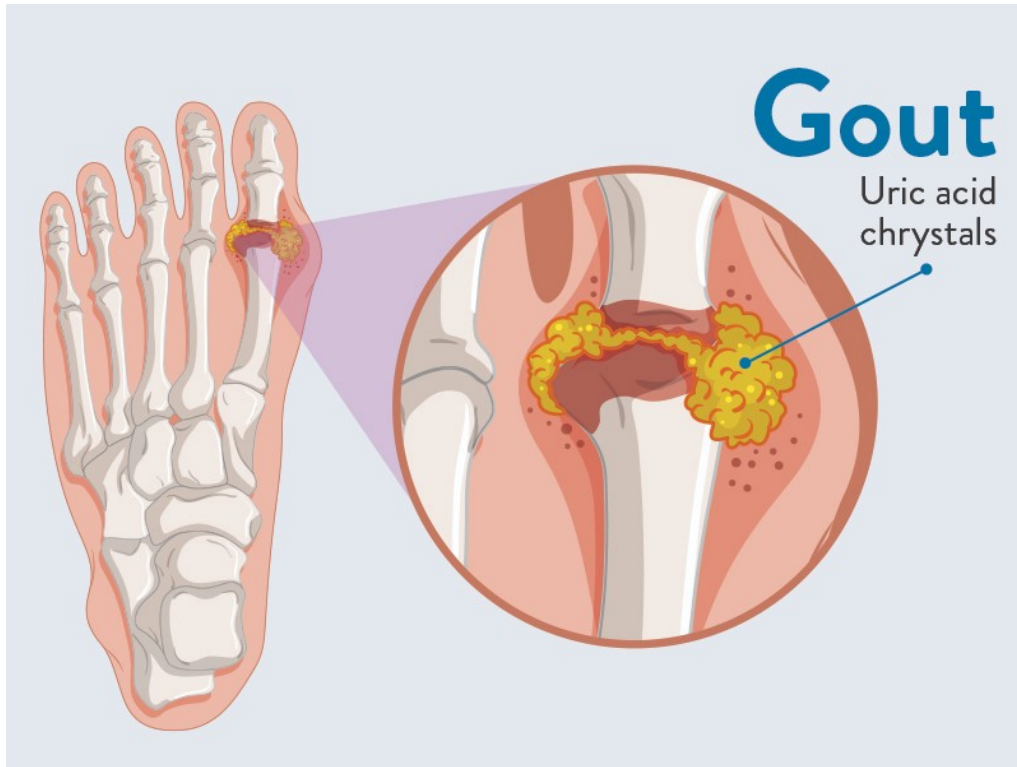
Protect from sun
exposure

Treatments for
impaired skin
integrity

Provide
psychosocial
support

GOUT

GOUT



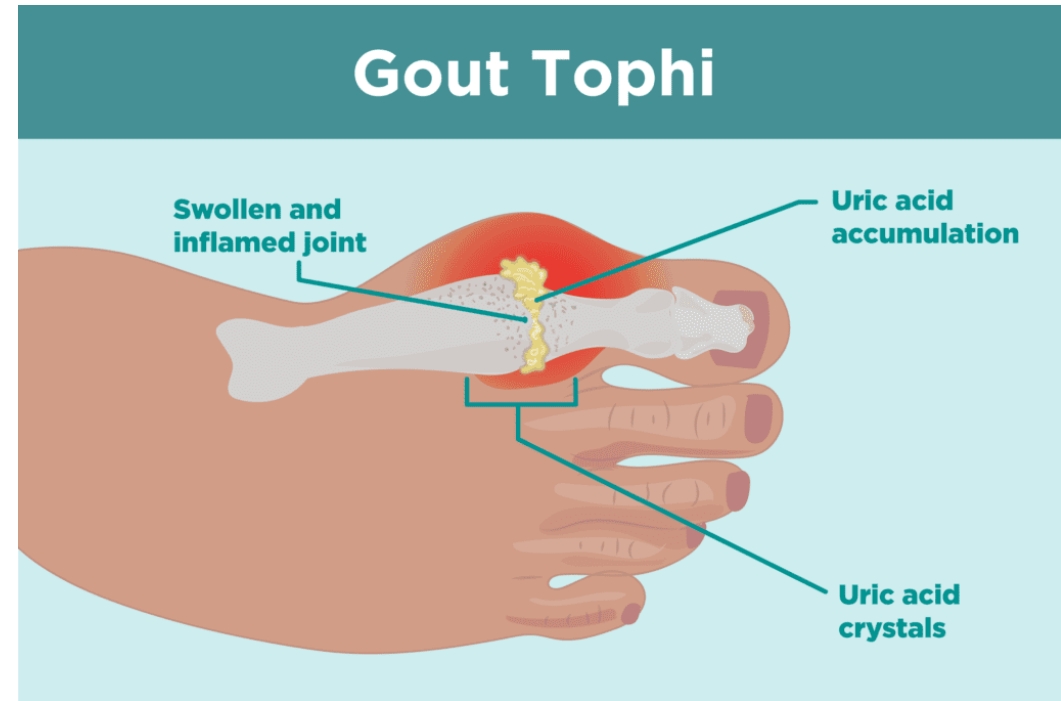
- Most common form of intermittent inflammatory arthritis
- Hyperuricemia occurs, causing uric acid crystals to deposit in joints and body tissues
- Uric acid results from the breakdown of purines in cells

GOUT: RISK FACTORS

- Increasing age
- Males are mainly affected until women reach menopause
 - Estrogen is protective against gout before menopause
- High BMI
- Genetic predisposition
- High intake of purine rich foods
- Alcohol ingestion
- Chronic renal failure
- Medications
 - Thiazide diuretics
 - Salicylates
 - Ethambutol
 - Nicotinic acid (niacin)
 - Cyclosporine

GOUT: CLINICAL MANIFESTATIONS

- Severe pain, redness, warmth, edema of affected joint
 - Most common in big toe joint
 - Very painful when the swollen joint is touched or moved
 - Appearance of tophi in soft tissues (chronic gout)
- Most attacks will subside over 3-10 days with treatment
- Attacks come and go, but result in gradual joint damage



GOUT: PHARMACOTHERAPY

- Medications are divided into 2 categories: preventive or treatment
- Preventive
 - Allopurinol
- Treatment
 - NSAIDs
 - Colchicine
 - Corticosteroids

What medication education should the nurse provide regarding pharmacotherapy for gout?

ANSWER: GOUT MEDICATION EDUCATION (1 OF 2)

- **Allopurinol (prevention)**

- Take every day, with food and 2–3 L of fluid daily; do not start during an acute attack, but do not stop it if one occurs while on it; report any rash immediately (severe hypersensitivity, especially in Asian clients – HLA-B*5801); regular uric acid, kidney and liver labs; takes weeks to lower uric acid, so a colchicine or NSAID “bridge” is often given at first.

- **Febuxostat, probenecid**

- Alternatives for prevention: febuxostat – watch for cardiovascular effects and liver labs; probenecid – lots of fluid, avoid aspirin (it blocks the drug).

- **NSAIDs (naproxen, indomethacin) – acute attack**

- First-line; start at the first twinge; take with food; avoid with kidney disease, ulcers or anticoagulants.

ANSWER: GOUT MEDICATION EDUCATION (2 OF 2)

- **Colchicine – acute attack**

- Start within 12–24 hours of onset; stop and call if diarrhea, nausea or vomiting start (toxicity); low dose in kidney disease; dangerous with statins, clarithromycin and grapefruit; keep out of reach of children.

- **Corticosteroids – acute attack**

- Oral, IM or injected into the joint when NSAIDs and colchicine cannot be used; short course; watch glucose.

- **For everyone**

- Avoid aspirin and thiazide diuretics; limit alcohol (beer and spirits), organ meats, shellfish, sugary drinks; lose weight slowly – crash diets trigger attacks; drink water; rest and elevate the joint during an attack.

GOUT: NURSING MANAGEMENT AND EDUCATION

- **Low-purine diet**
- Limit intake of ETOH
- Avoid starvation diets, aspirin, and diuretics
- Limit physical or emotional stress
- Increase fluid intake
- Maintain normal body weight

Low-Purine Diet for Gout

Eat More



Fruits



Vegetables



Whole grains



Low-fat dairy



Legumes



Nuts

Eat Less



Shellfish



Organ meats (liver)



Alcoholic beverages



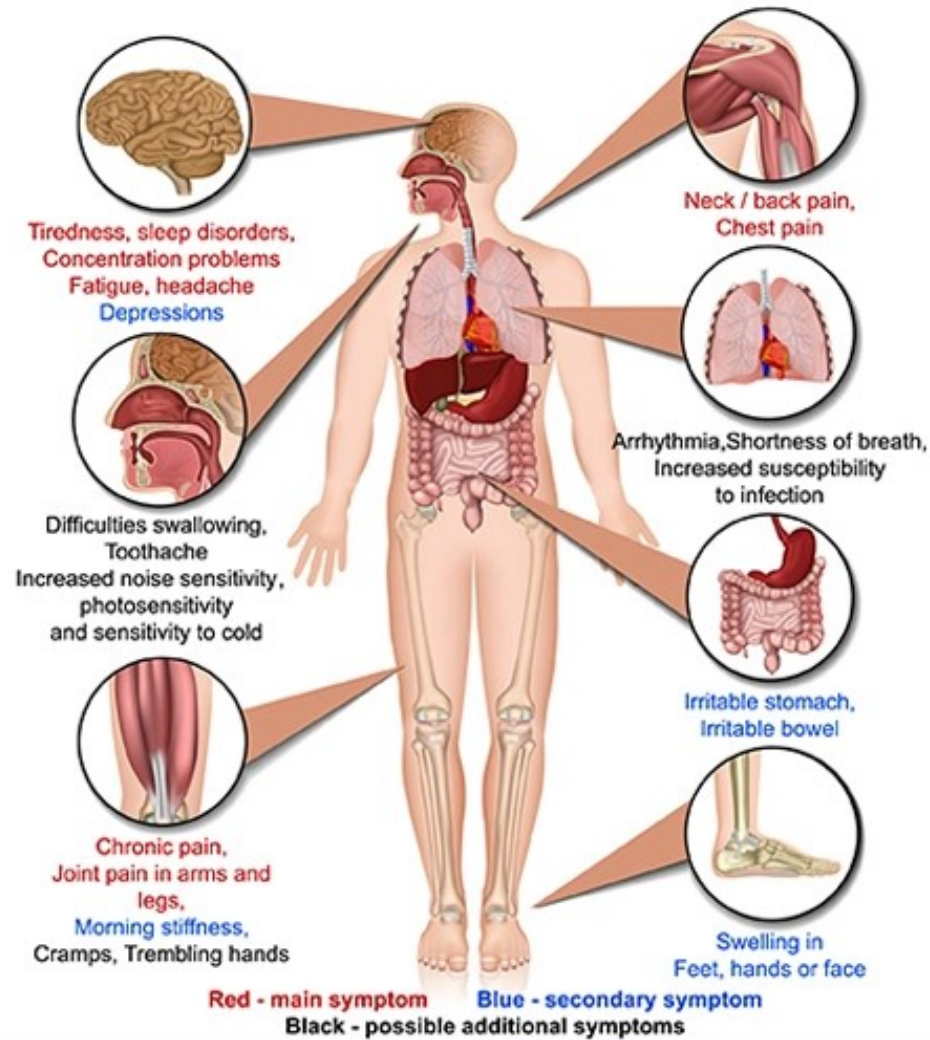
Soft drinks

FIBROMYALGIA

FIBROMYALGIA

- Chronic disorder that causes extensive musculoskeletal pain, fatigue, and sleep, cognitive, and mood problems.
- Pain sensations are increased related to altered brain and spinal cord interpretation of painful and non-painful signals.
- Symptoms usually appear after a traumatic event, such as surgery, infection, or considerable psychological stress.
- Tension headaches, temporomandibular joint (TMJ) issues, irritable bowel syndrome, anxiety, and depression are common manifestations
- More common in women

FIBROMYALGIA SYMPTOMS



FIBROMYALGIA : DIAGNOSTICS AND MEDICAL MANAGEMENT

- No specific tests for fibromyalgia
- H&P
- Often diagnosed through a “process of elimination.”
- Medical treatment is aimed at reducing symptoms.
- Pharmacotherapy
 - NSAIDs
 - SNRIs
 - GABA analogs
 - TCAs or zolpidem
- Exercise, activity, PT/OT
- Sleep hygiene

FIBROMYALGIA: NURSING MANAGEMENT AND EDUCATION

- Assess/monitor pain, fatigue, mobility
- Assist with symptom management
 - Limit intake of caffeine, alcohol, and other substances that interfere with sleep
 - Develop a sleep routine
 - Engage in low-impact exercise daily
 - Acupuncture, stress, management, tai chi, hypnosis can also be used
 - Medication education
- Provide support and encouragement
 - Clients typically have had symptoms for extensive period and may feel that their symptoms have not been taken seriously